$$w = \frac{\mathrm{v} + \mathrm{v}'}{1 + \frac{\mathrm{v}\mathrm{v}'}{c^2}} \ . \tag{16}$$

$$dw = \frac{\left(1 - \frac{\mathrm{v}^2}{c^2}\right)}{\left(1 + \frac{\mathrm{v}\mathrm{v}'}{c^2}\right)^2} d\mathrm{v}' . \tag{17}$$

$$t = \frac{t' + \frac{\mathrm{v}x'}{c^2}}{\sqrt{1 - \mathrm{v}^2 / c^2}} \tag{18}$$

$$dt = dt' \frac{\left(1 + \frac{\mathrm{v}\mathrm{v}'}{c^2}\right)}{\sqrt{1 - \mathrm{v}^2 / c^2}} . \tag{19}$$

(17) (19) $\mathrm{v}' = 0$

$$a_r = \frac{dw}{dt} = \left(1 - \frac{\mathrm{v}^2}{c^2}\right)^{3/2} \frac{d\mathrm{v}'}{dt'} = \left(1 - \frac{\mathrm{v}^2}{c^2}\right)^{3/2} a_p . \tag{20}$$

8. ,

$$a_p = \frac{d\mathrm{v}'}{dt'} = \frac{u}{m} \frac{dm}{dt'} . \tag{21}$$

(20) (19) $\mathrm{v}' = 0$

$$\frac{dm}{dv} = \frac{m}{u(1 - \mathrm{v}^2 / c^2)} . \tag{22}$$

$$\alpha = \frac{c}{2u} . \tag{23}$$

9.

$$m_0 = m\left(\frac{1 + \mathrm{v}/c}{1 - \mathrm{v}/c}\right)^{c/2u} = 10^{28630}\, kg. \tag{24}$$

10.

$$m_0 = m\left(\frac{1 + \mathrm{v}/c}{1 - \mathrm{v}/c}\right)^{c/2u} = 1730\, kg. \tag{25}$$

채점 기준

№	평가 항목	배점	
1	(1)	0.25	**0.5**
	(2)	0.25	
2	(3)	0.25	**0.5**
	(4)	0.25	
3	(5)	0.25	
	(6)	0.25	**0.75**
	(7)	0.25	